Engineering Chemistry · Unit_01 · Molecular spectroscopy, phase equilibria, and computational chemistry

# DRAFT — UE26CY151A · Unit_01 · Molecular spectroscopy, phase equilibria, and computational chemistry · Electronic spectroscopy - Franck-Condon principle (Part II)

Standalone video lesson · authored study notes with source citations

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# · How to use this pack

Read the topic explanation, worked examples, and recap. Evidence citations point to source transcript intervals and slide pages; full slide images remain in the source visual index. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

| | |
|---|---|
| Source text | Verbatim or OCR-derived text, kept separate from editorial notes. |
| Candidate | Automated topic/source association requiring manual confirmation. |
| Low ASR confidence | Transcript segment requires audio verification. |
| DRAFT | Human review remains incomplete. |

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M1

# Electronic spectroscopy - Franck-Condon principle (Part II)

## 1 Electronic spectroscopy - Franck-Condon principle (Part II)

Study focus: Electronic spectroscopy - Franck-Condon principle (Part II). This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment.

EV0000110 · EV0000111

SHORT-NOTE RECAP

**Core explanation · 00:00–04:59**

We will continue with module one on molecular spectroscopy.

**Key details**

- This will be the last class of this module.
- Today's class we are going to discuss about the intensity of spectral lines and the Frank Condon principle.
- We are studying electronic spectroscopy.
- Which means that when we allow UV visible light to fall on the molecule, the electron can undergo transition from the ground state to the excited state.
- And within the electronic state we know that there are vibrational level and these vibration levels you can see are.
- Coming closer and closer, the energy gap between the levels is coming, is becoming less, which shows that the molecule is an harmonic in nature.
- Also, the energy profile is a molecular.
- curve that shows us that the molecule is undergoing anharmonic oscillations and V double prime equal to 01234 is representing the vibrational levels in the ground state and v prime equal to 0123 represents the vibration levels in the excited state.
- It means that when an electron undergoes transition from the ground state to the excited state, it can end up in any of these vibrational levels.
- There is no rule forbidding any vibrational change.
- Vibrational levels of the excited state.

- So that means the spectrum is going to be very complicated.
- but we have a rule which helps us simplify the spectrum that most of the transitions start from V double prime equal to zero as it is the most populated level.
- And that becomes the most populated level.
- And therefore, most of the transitions will start from V double prime equal to zero.
- These vibration levels in the excited state, that's because of the rule one.
- Now the third rule says the excited state usually has longer internuclear distance because of the anti-bonding character.
- And if you look at the internuclear distance of the excited state, we see that it is definitely larger.
- So if the antibonding character is more, the number of antibonding electrons are more and therefore we get a small bond order.
- And we know that bond order and bond length are inversely proportional.
- So if the bond order is small, that means the bond length is going to be more.
- The next rule says the transitions arise from the center of V double prime equal to 0 because the maximum probability of finding the nuclei is at the center of V double prime equal to 0.
- They should be actually V double prime equal to zero, V double prime equal to one, two, three, four.
- But because we have shown only one electron state, we have written it as V is equal to zero, one, two, three.
- But since we are talking of electronic transitions.

**Example or application from the lecture**

- We know that bond order from molecular orbital theory we know that bond order is given by number of bonding electrons minus the number of antibonding electrons divided by two.

*These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.*

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**Core explanation · 05:00–09:59**

we should denote the ground state as V double prime equal to zero.

**Key details**

- The psi squared becomes the probability of finding the nuclear.
- For V is equal to 0, we find that this is the shape.
- And here we see that in the center there is maximum probability of finding the nuclei.
- As we go to V is equal to 1, we find that it becomes like this.
- And then as we go to higher and higher values of V, we find that the probability of finding the nuclei is more at the extremity as for V is equal to zero, the probability of finding the electrons in is more in the center.
- And therefore that is going to help us further understand the pattern.
- V double prime equal to zero is the one which is most populated level at room temperature and.

- Here from the probability graph we can understand that the center is where we are going to have the electrons.
- It is not that there are no electrons here or the nuclei do not exist in these positions at any point of time, but more probably of finding the nuclei is at the center for V double prime equal to zero.
- Rotational spectroscopy, vibrational spectroscopy, as well as electronic spectroscopy and the rules which are governing the energy profile and the placement of these.
- s energy levels.
- So for both ground as well as the excited state and therefore we know that it is.
- And harmonic in nature, and also the vibrational levels seem to be coming closer, and that also is an indication that we are talking of anharmonic anharmonically oscillating molecule.
- The ground state we have these vibrational levels, and within the excited state also we have these vibrational lepths.
- Within these vibrational levels we have the rotational levels.
- J value, then we find that these rotational levels become further and further apart.
- And that is what we have understood.
- Within each of these electronic levels we have the vibrational levels, and within the vibrational levels we have this rotational levels.
- If I want an electronic transition to take place, or if I want the molecule to move from the ground electronic state to the higelectronic state, it requires a lot of energy and that energy is corresponding to UV visible light.
- Whereas if I want just a vibrational change in the molecule, then this small amount of energy is what is required and that happens to be in the infrared region.
- why we draw the energy level diagram this way and also we know that because of the anti-bonding character there is a slight displacement in the excited state.

**Example or application from the lecture**

- If you remember in the initial classes I had given this diagram, but we did not clearly understand why we are drawing the diagram in this way for a diatomic molecule.
- Let us now look at Frank Cordon principle, which is going to explain the intensity of the spectrum.

*These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.*

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**Core explanation · 10:00–14:59**

So that means all lines should be of equal intensity, but it is not found to be so.

**Key details**

- That is explained with the help of Frank Contin principle, which states that an electronic transition takes place so rapidly that a vibrating molecule does not change its internuclear distance appreciably during the transition.
- the electron being very light undergoes transition very quickly and the nuclei being bulky they take time to adjust to this transition and therefore for the electron it is as if the nuclei

are stationary.
- That in this case, what you see here that the electron has moved from V double prime equal to 0 to V prime equal to 2.
- And therefore, since.
- V double prime equal to zero is the state which is most populated and at the same time we know that transitions will happen from the center of V double prime equal to zero.
- So most of the transitions are happening from happening from here and therefore two zero line will be most intense in this.
- RE double prime and r e prime are the internuclear distances of the ground and excited states respectively.
- And RE double prime would correspond to the internuclear distance for the ground state whereas R prime would stand for the internuclear distance of the excited state.
- The first one states that the internuclear distance of the excited state is more than that of the ground state.
- Double prime equal to zero is the most populated state and transitions most of the transitions will happen from the center because that is the where we have the maximum probability of finding the nuclear.
- So therefore in this case we find that.
- Two zero is the line with maximum intensity, but that doesn't mean we don't have other lines, there are other lines also, but the probability of finding the nucleide electron is much less.
- in these areas and therefore we find that there are lines corresponding to 1000 and also three zero four zero etc but though those intensities are much less as compared to two zero and here we see that two zero line is here and this is centimeter inverse meaning that energy is increasing in this way one zero definitely would be to the left of two zero.
- Because the position represents the energy gap.
- And therefore we find that one zero is less than in energy so we will show the line here but nevertheless the intensity is less because the intensity is always more in the center of e double prime equal to 0 and similarly 0 will be to the left 30 though the intensity is less it is appearing to the right of 2 0.
- Another case would be when there is no change in the internuclear distance.
- In that case, again there is going to be a vertical transition and we find that.
- It happens from the center and ends up in the center of V prime equal to 0.
- And therefore the maximum intensity now will be for the 00 line.
- And there are other possibilities that the probability of finding the new.

**Example or application from the lecture**

- Let us study this in little more detail.
- Let us see what they are.

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**Core explanation · 15:00–19:59**

Does exist on either side and therefore we find one zero two zero three zero lines are also seen in the spectrum.

**Key details**

- Nevertheless, one zero will be to the right of zero because zero requires the minimum energy, one zero will require little more energy.
- And since the spectrum showing the energy scale in this direction, therefore we find that one zero will be here.
- then in that case zero nine is that it has the maximum intensity.
- The third case would be where the internuclear distance of the excited state is considerably greater than that of the ground state.
- So that means RE double prime is much less than RE prime.
- You can see that it is very much displaced to the right.
- Dissociation energy of the excited state.
- These atoms gain kinetic energy, and now the energy is not quantized anymore.
- This kinetic energy.
- Energy is not quantized, and when energy is not quantized, we cannot get discrete lines, and what we see is a continuum.
- You can see in the spectrum here, we see a continuum in the spectrum.
- Nevertheless, there are some lines to the left of the continuum.
- And those are of lesser intensity because there is probability of finding the nuclei around the center also.
- But nevertheless, the maximum intensity is from the la for the line which is originating from the center.
- But since gone beyond the dissociation energy, what we see is a continuum in the spectrum.
- That brings us to the end of module one.
- In the next class, we are going to talk about nanomaterials.

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EV0000122 · EV0000123 · EV0000124 · +1 source units in pack.json

# A Symbol glossary

| Symbol | Name | Meaning | Appears in |
|---|---|---|---|

# B Sources, coverage and gaps

| Source | Path | Use |
|---|---|---|
| Electronic spectroscopy - Franck-Condon principle (Part II) transcript (JSON) | https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/AV_Summaries/007_Electronic_spectroscopy_-_Franck-Condon_principle_Part_II__70e8e6f497126f7d.transcript.json | Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps. |
| Electronic spectroscopy - Franck-Condon principle (Part II) transcript (Markdown) | https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/AV_Summaries/007_Electronic_spectroscopy_-_Franck-Condon_principle_Part_II__70e8e6f497126f7d.transcript.md | Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps. |
| Electronic spectroscopy - Franck-Condon principle (Part II) transcript (plain text) | https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/AV_Summaries/007_Electronic_spectroscopy_-_Franck-Condon_principle_Part_II__70e8e6f497126f7d.transcript.txt | Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9987; only the JSON segmentation supplies the displayed timestamps. |
| Electronic spectroscopy - Franck-Condon principle (Part II) transcript (SRT) | https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/AV_Summaries/007_Electronic_spectroscopy_-_Franck-Condon_principle_Part_II__70e8e6f497126f7d.srt | Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps. |

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

### Known gaps, conflicts and corrections

1. 2 transcript segments have low ASR confidence; they are flagged with timestamps in the transcript blocks and indexed in the corpus.
2. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.
3. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
4. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.

5. The study PDF contains edited, topic-organized lecture notes and textual slide-page takeaways. Full-page slide images remain in the source/visual index datasets, not embedded as the lesson. Human content, ASR, formula, and page-level visual review remains pending.

## Coverage

Evidence is summarized by source to keep the reading copy legible. Full source-unit IDs, locators and decisions remain in pack.json.

| Source | Unit counts | Example decisions |
|---|---|---|
| SRC00022 | 4 covered · 0 excluded · 0 gaps | t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare...<br>t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0043. Sidecar variants compare...<br>t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0044 to s0064. Sidecar variants compare... |
| SRC00023 | 4 covered · 0 excluded · 0 gaps | t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare...<br>t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0043. Sidecar variants compare...<br>t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0044 to s0064. Sidecar variants compare... |

| Source | Unit counts | Example decisions |
|---|---|---|
| SRC00024 | 4 covered · 0 excluded · 0 gaps | t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare...<br>t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0043. Sidecar variants compare...<br>t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0044 to s0064. Sidecar variants compare... |
| SRC00025 | 4 covered · 0 excluded · 0 gaps | t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare...<br>t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0043. Sidecar variants compare...<br>t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0044 to s0064. Sidecar variants compare... |

# C Revision and retrieval practice